TPC-414: Three-Shell Pooled Extension

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Abstract

We add the complete shell at $Q = 262144$ to the pooled two-shell proxy. The three shells retain $5709 + 10749 + 20390 = 36848$ primes. With shell-local amplitudes, $H = 66$, and $N = 264$, the pooled alternating CRT profile has $m_- = m_+ = 18424$. Exact rational arithmetic and a fresh literal replay verify the resulting finite row; no arithmetic theorem is claimed.

1 Three-shell profile

Let $\mathcal{P}$ be the increasing union of $(65536, 131072]$, $(131072, 262144]$, and $(262144, 524288]$. For $p_i \in \mathcal{P}$, let Q_i be its source-shell scale and put $a_i = p_i^3/[Q_i^2(p_i - 1)]$. Set $H = 66$, $N = 264 = 4H$, and use CRT residues 0 on even indices and $-N$ on odd indices. The pooled count is 36848, so $m_- = m_+ = 18424$. Every prime exceeds N; hence

$$G_0 = V_- S_0, \qquad G_1 = V_- S_1 + V_+(S_1 - t_1^2), \qquad M = t_1 P_-.$$

Since $P_-^2 \leq m_- V_-$, $V_- \geq m_- a_{\min}^2$, $G_1 \geq V_- S_1$, and $S_0, S_1 \geq H/4$,

$$0 \leq z \leq \frac{t_1}{a_{\min}\sqrt{S_0 S_1}} \leq \frac{4}{a_{\min} H} \leq \frac{4}{H}. \tag{1}$$

2 Exact observation

The certificate contains all 36848 prime labels, their shell scales, the CRT residue and exact rational energies. Its decimal normalized observation is a finite numerical value, not an asymptotic claim. The independent checker rebuilds the three sieves and literally visits every per-prime/per-coordinate mask before comparing G_0, G_1, and M.

3 Scope and reproduction

This one finite synthetic adjacent normalized proxy entry does not prove a full operator estimate, physical h_0 or arithmetic signs, pay arithmetic L^2 or fixed-power credit, close Route A/B, or imply a twin-prime result. Run the producer, replay, and mutation stress with `--check` in normal and optimized modes; Bridge-B repeats and locks these checks.